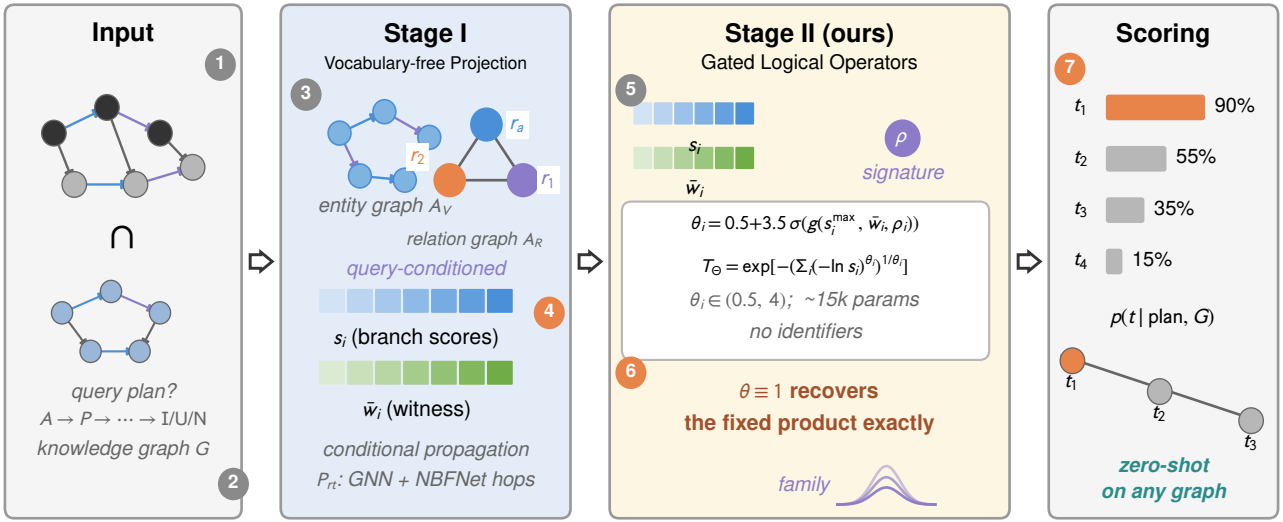

Input
1
query plan?
A → P → ⋯ → I/U/N
knowledge graph G
2
Stage I
Vocabulary-free Projection
3
entity graph A_V
relation graph A_R
query-conditioned
4
s_i (branch scores)
$\bar{w}_i$ (witness)
conditional propagation
P_{rt}: GNN + NBFNet hops
Stage II (ours)
Gated Logical Operators
5
s_i
$\bar{w}_i$
signature
$\theta_i = 0.5+3.5\,\sigma(g(s_i^{\max}, \bar{w}_i, \rho_i))$
$T_\Theta = \exp[-(\Sigma_i(-\ln s_i)^{\theta_i})^{1/\theta_i}]$
$\theta_i \in (0.5, 4)$; ~15k params
no identifiers
6
$\theta \equiv 1$ recovers
the fixed product exactly
family
Scoring
7
t_1 90%
t_2 55%
t_3 35%
t_4 15%
$p(t \mid \text{plan}, G)$
zero-shot
on any graph